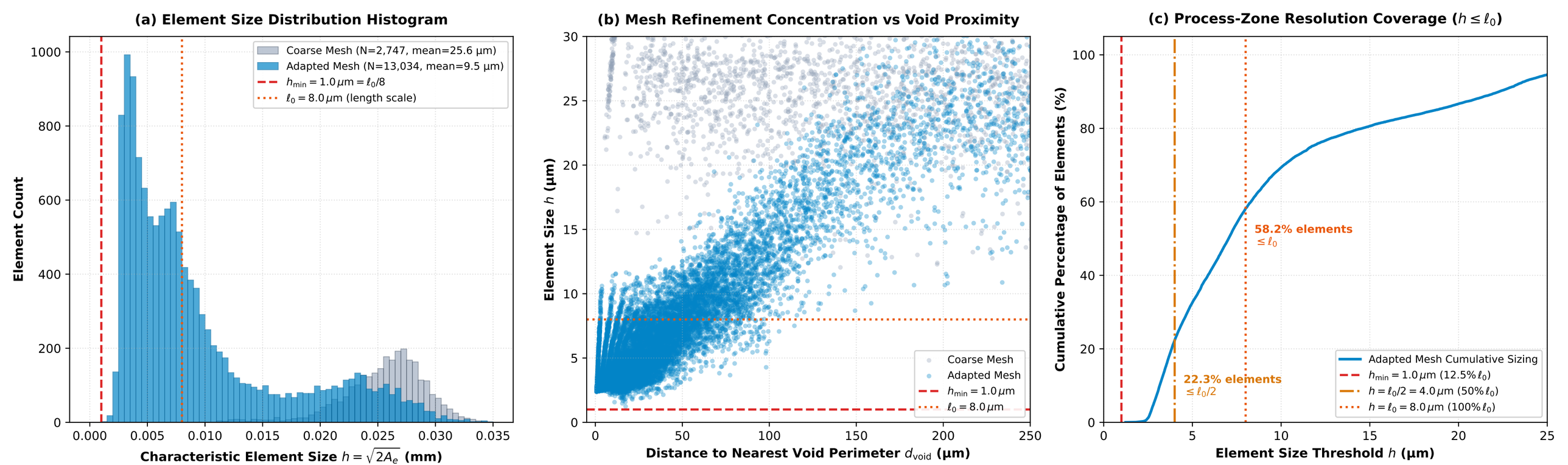

(a) Element Size Distribution Histogram
Coarse Mesh (N=2,747, mean=25.6 μm)
Adapted Mesh (N=13,034, mean=9.5 μm)
$h_{min} = 1.0\,\mu m = \ell_0/8$
$\ell_0 = 8.0\,\mu m$ (length scale)
Element Count
Characteristic Element Size $h = \sqrt{2A_e}$ (mm)
(b) Mesh Refinement Concentration vs Void Proximity
Element Size h (μm)
Distance to Nearest Void Perimeter d_{void} (μm)
Coarse Mesh
Adapted Mesh
$h_{min} = 1.0\,\mu m$
$\ell_0 = 8.0\,\mu m$
(c) Process-Zone Resolution Coverage ($h \le \ell_0$)
Cumulative Percentage of Elements (%)
Element Size Threshold h (μm)
58.2% elements
$\le \ell_0$
22.3% elements
$\le \ell_0/2$
Adapted Mesh Cumulative Sizing
$h_{min} = 1.0\,\mu m$ (12.5% ℓ_0)
$h = \ell_0/2 = 4.0\,\mu m$ (50% ℓ_0)
$h = \ell_0 = 8.0\,\mu m$ (100% ℓ_0)